

AI LITERACY WORKSHOP BY



Intro to **Phyiscal AI** with Edge AI Vision Demo Project

PRESENTED BY AIOSCOUT

Agenda

01

1 · Intro lecture

~15 min

- Intro to physical AI components
- Model training terms explanation
- Brief intro to physical AI ethics

02

2 · Lab tutorial

~15 min

Introduce to our training platform and watch the instructor train a road-sign recognizer live.

03

3 · Hands-on lab

~60 min

In teams of 3: build own digit recognizer (0–9) and run it on the car.

04

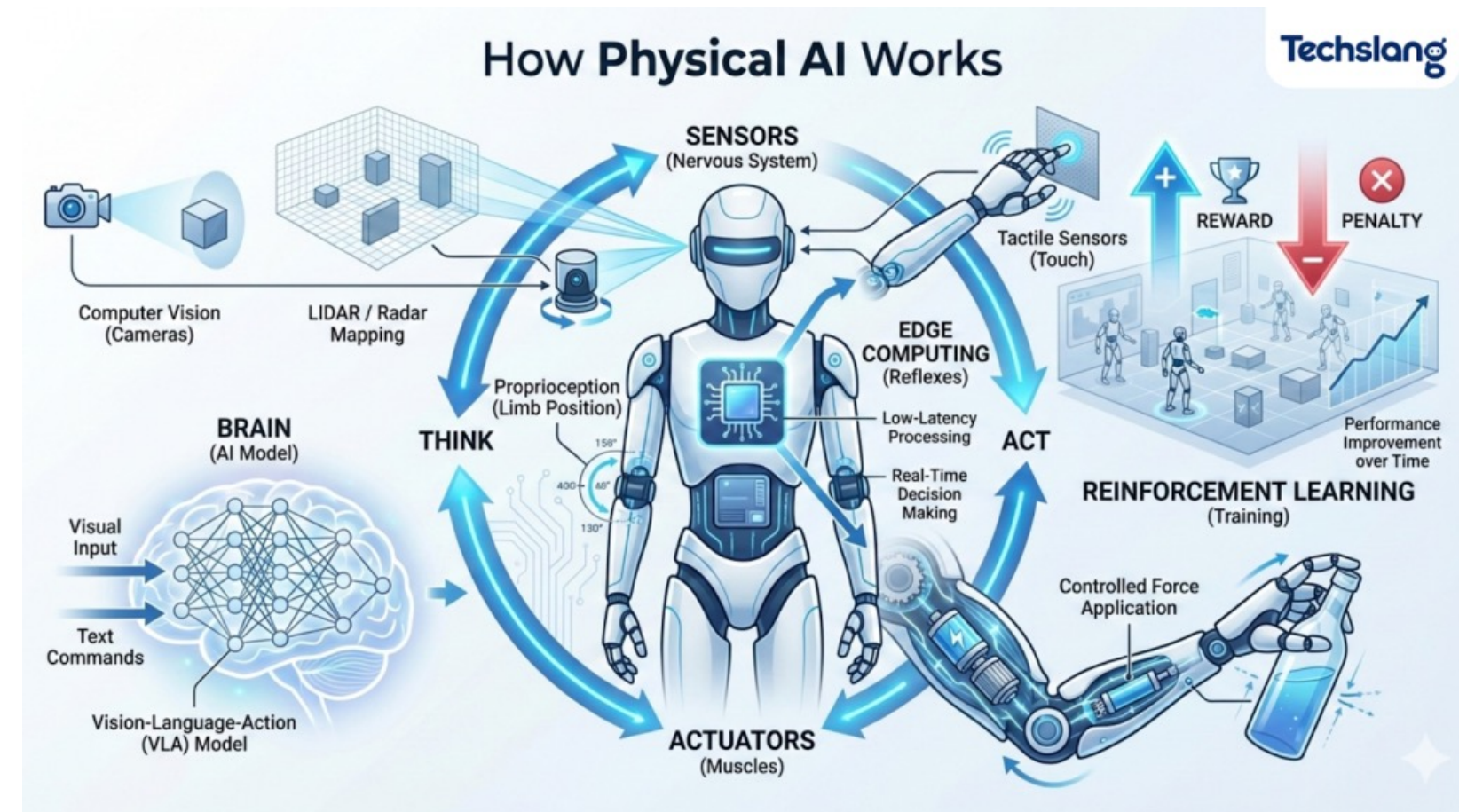
4 · Demo & sharing

~30 min

Each team demos their model, shows their score, shares one lesson.

What is Physical AI?

Chatbot AI is what you may already be familiar with, but Physical AI is very different, and have greater potential on doing much more different tasks



You already meet Physical AI



Robot vacuum

feels walls, avoids stairs, goes home to charge



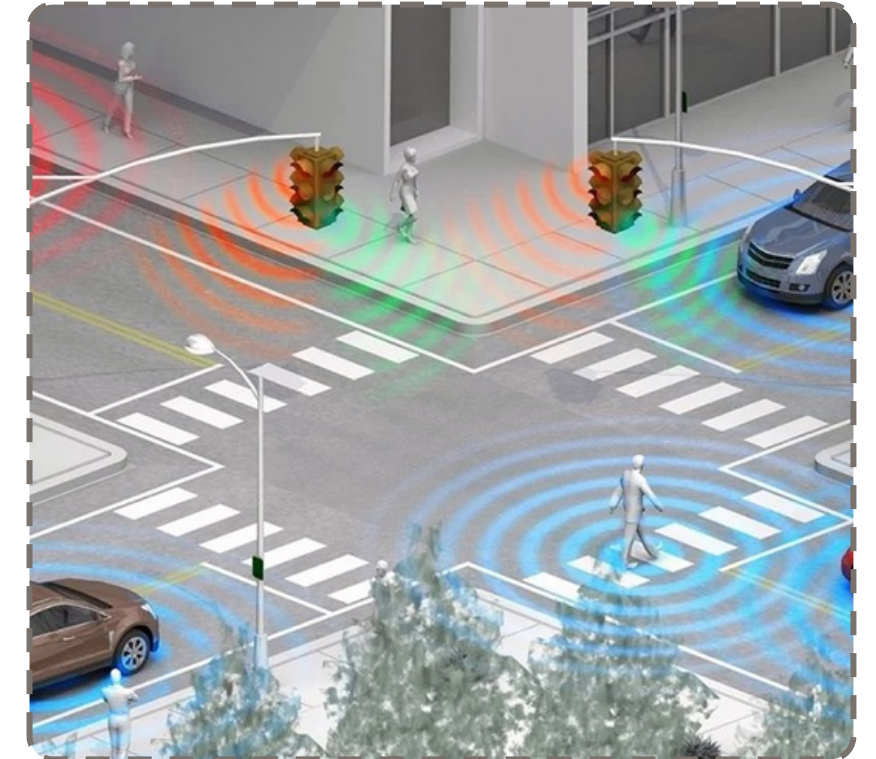
Delivery drone

finds the landing spot with its camera



Self-driving car

reads lanes, signs, and people — like ours, but giant



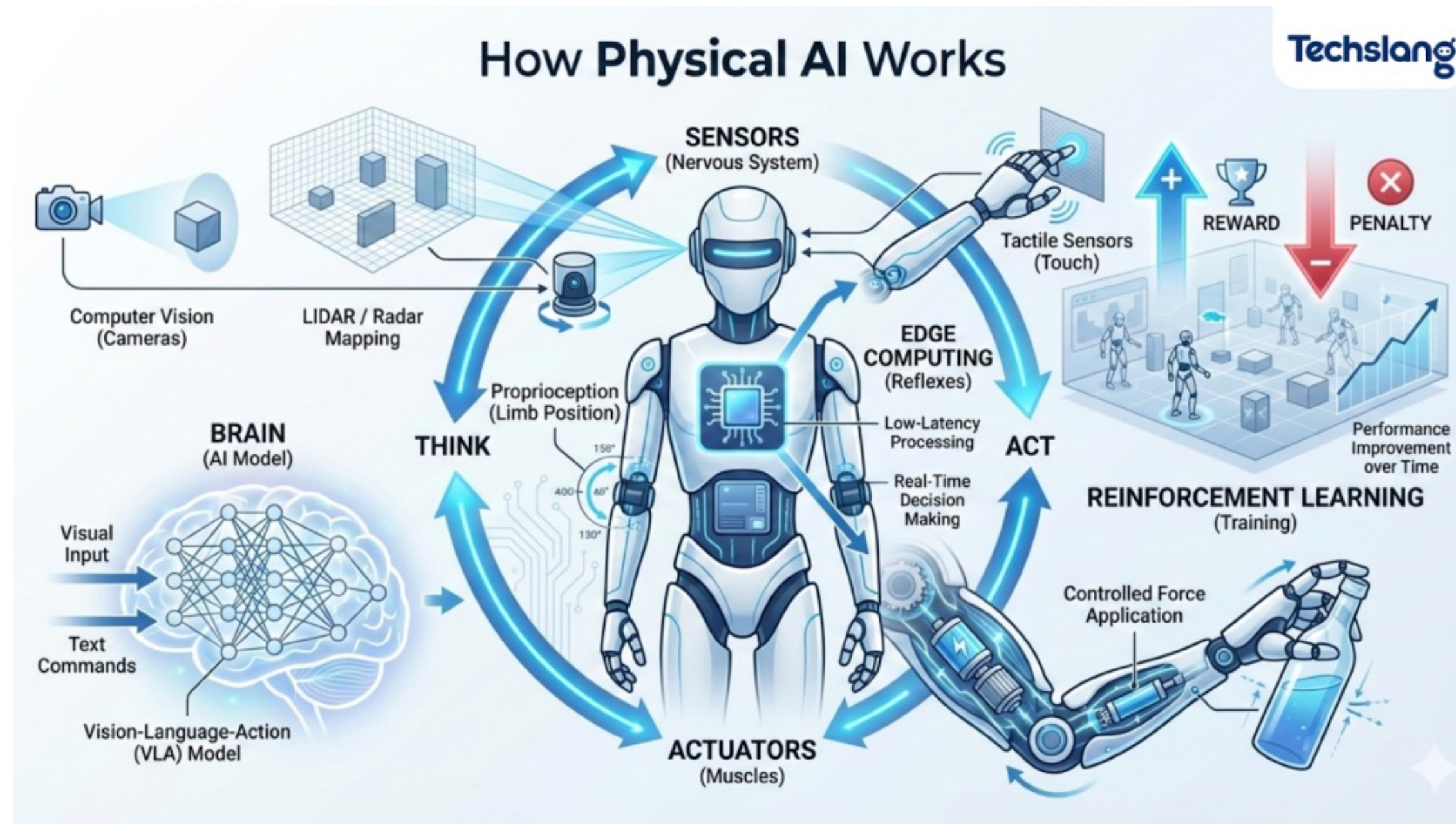
Smart traffic light

watches the road and changes its own timing

Let's now look at some import parts of Physical AI

What is Physical AI?

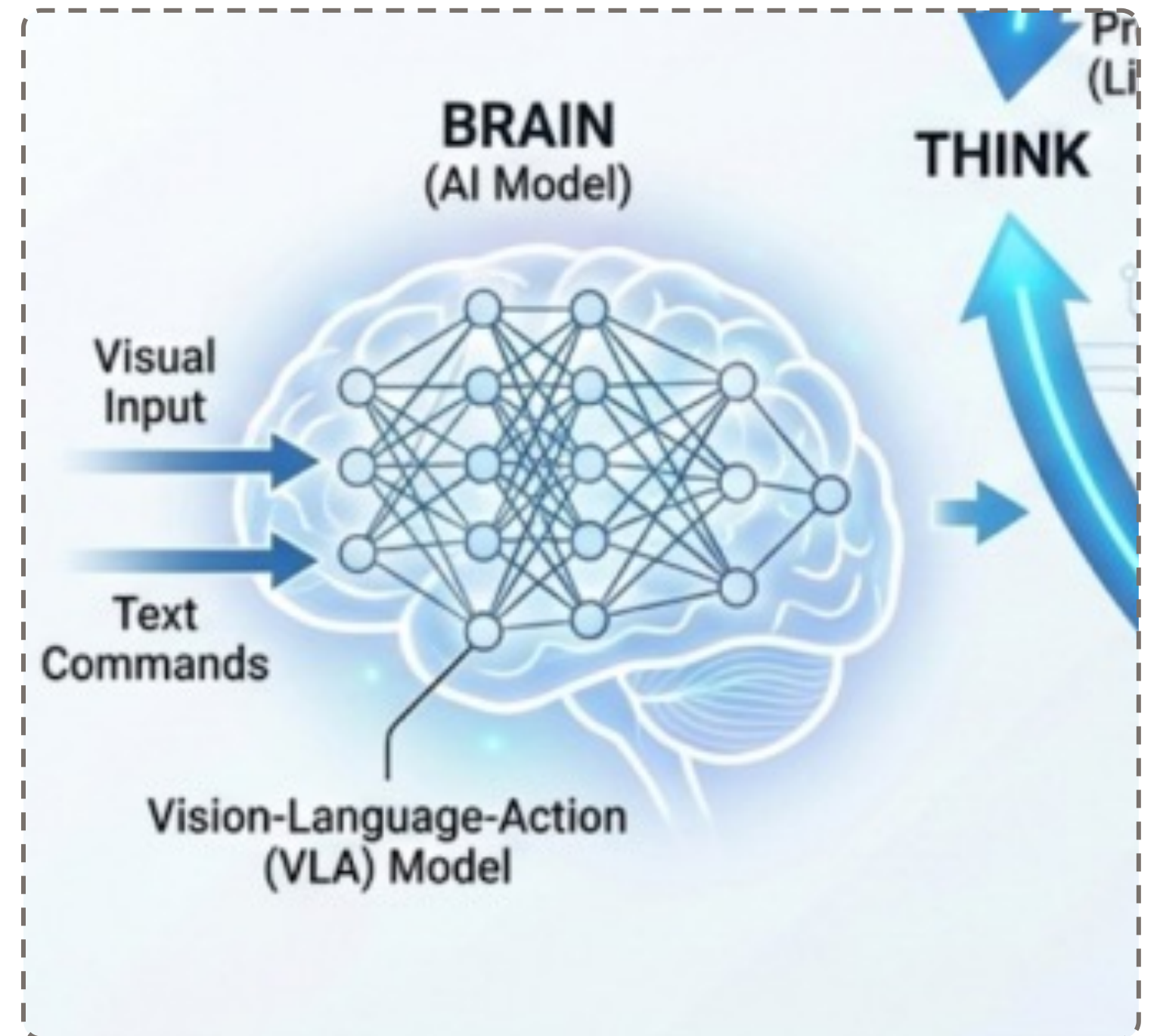
Could you name any familiar components?



AI Model

≈ the brain

- A program that has learned (trained) from examples
- It looks at what the sensors see and decides what to do next

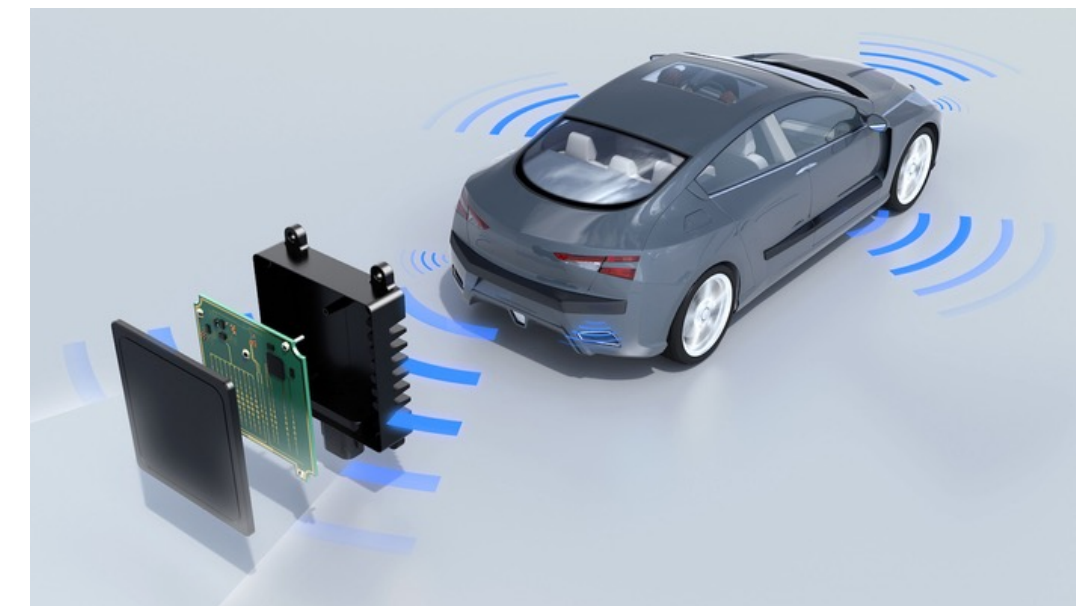
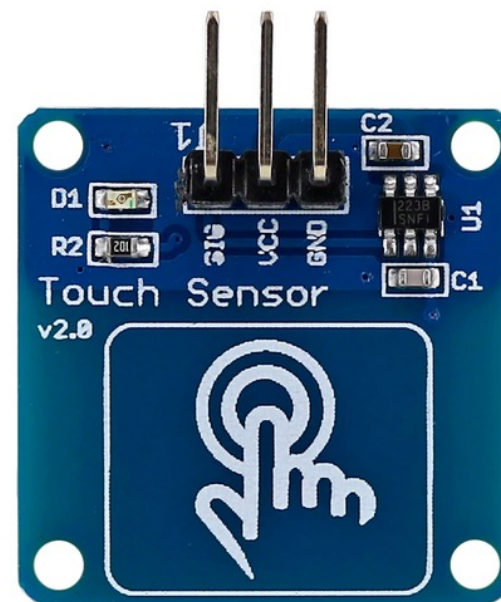
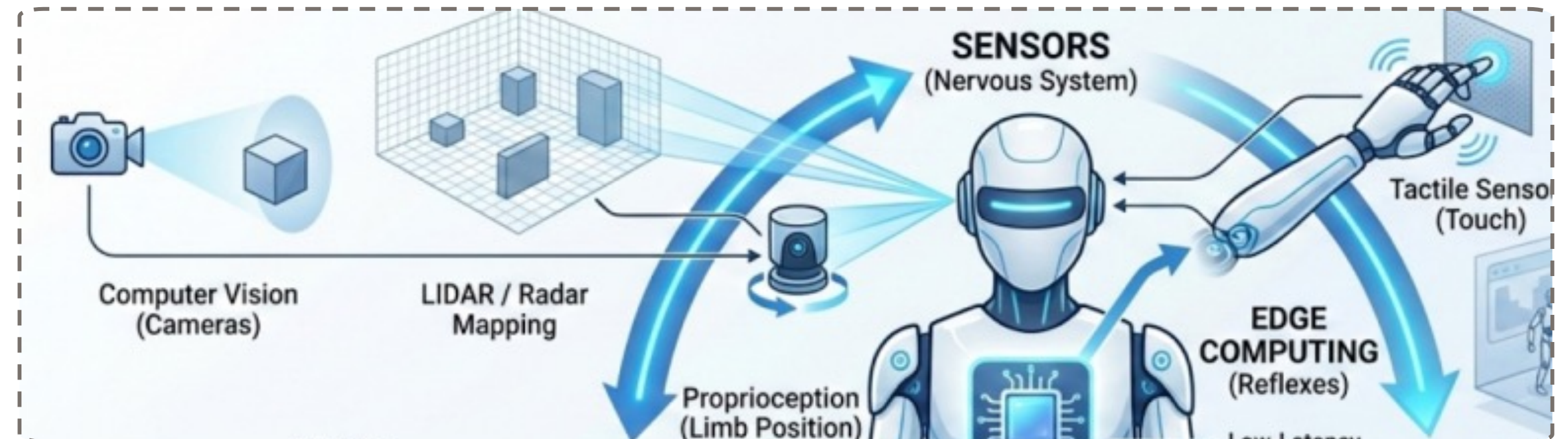


Sensors - nervous systems

e.g. hands / eyes / nose

Devices that measure the real world,
for example:

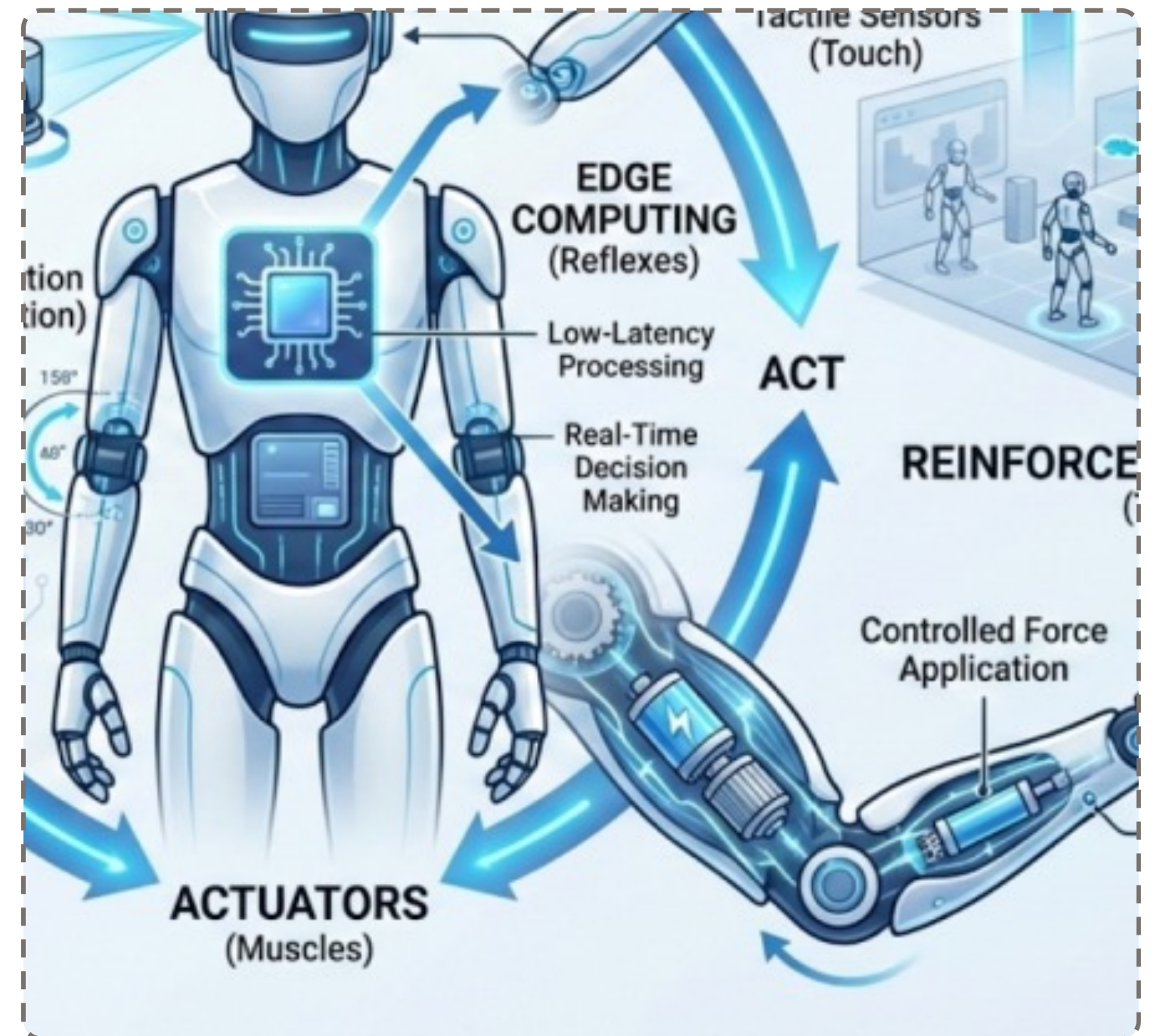
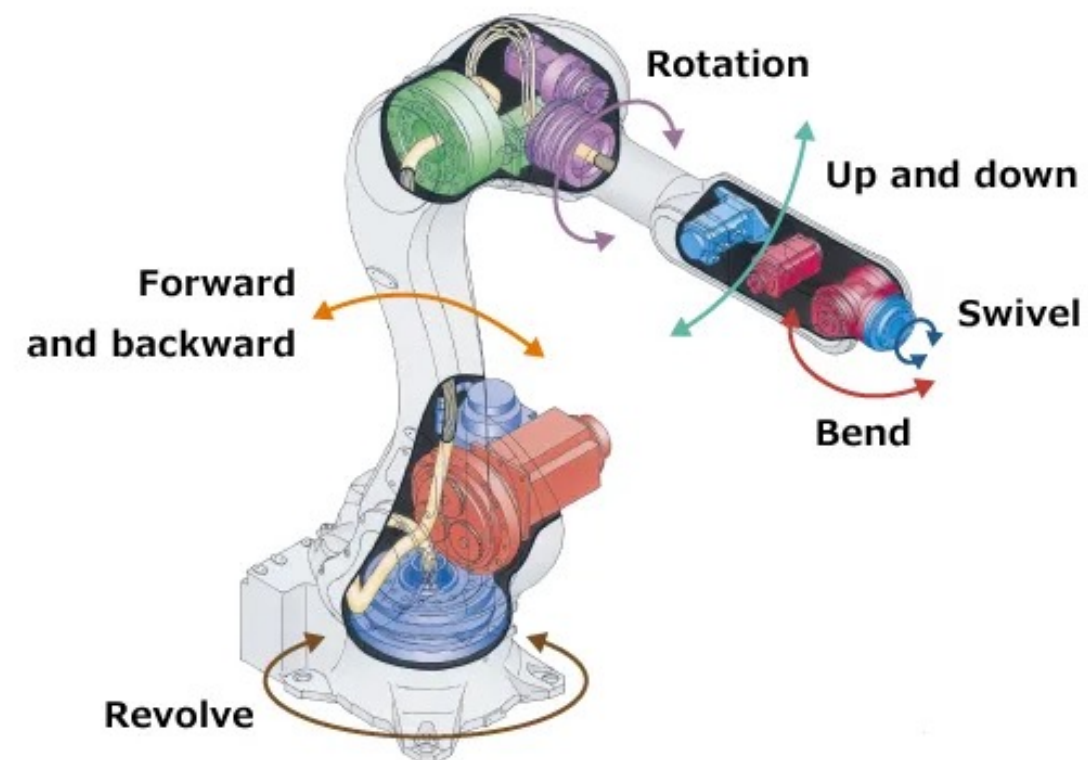
- Cameras see
- Touch sensors feel
- Radar measures distance



Actuators

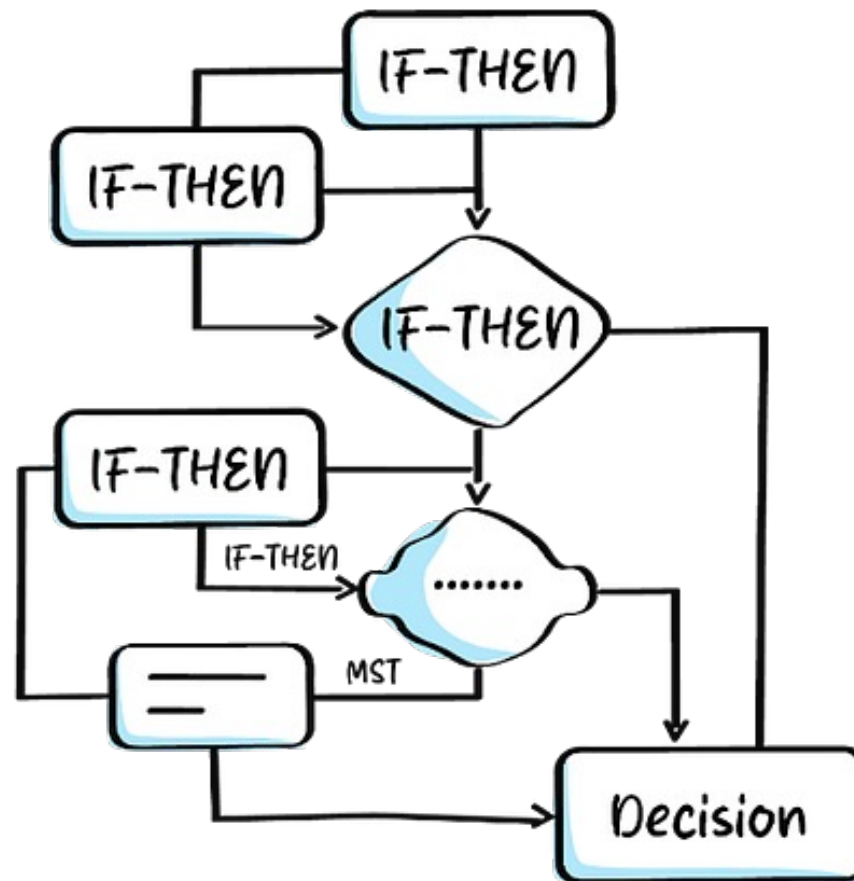
≈ Muscles

- Parts that move the body, for example motors spin wheels, joints move arms.
- They turn the AI's decision into action.

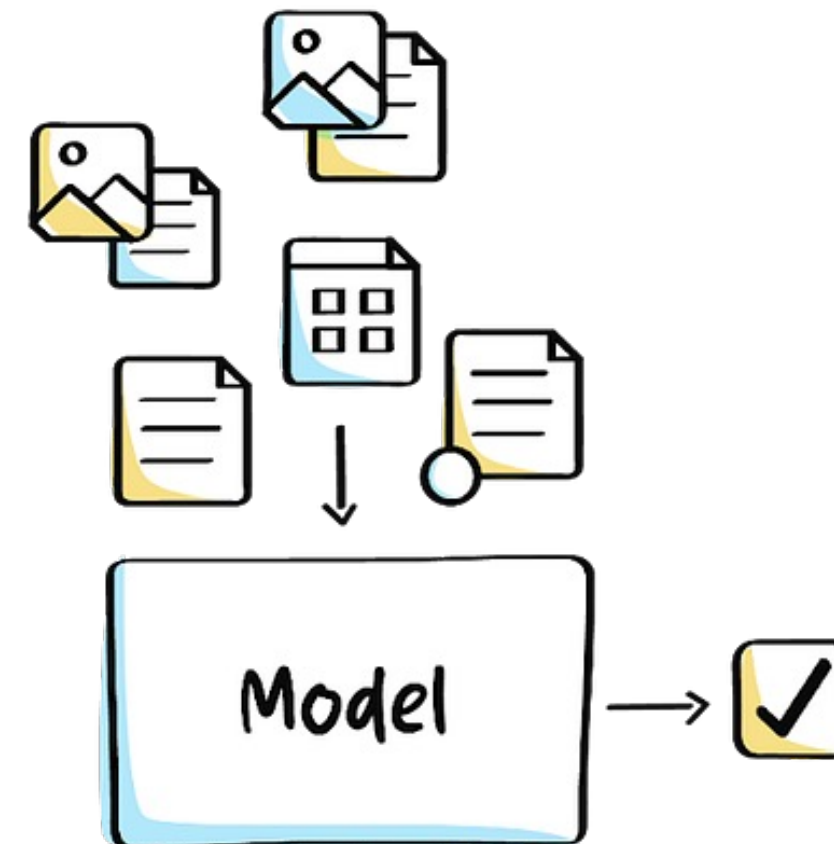


How is the model trained?

Rule-Based System



ML-Based System



How is the model trained?

WAY 1 · RULE-BASED

A human writes every rule

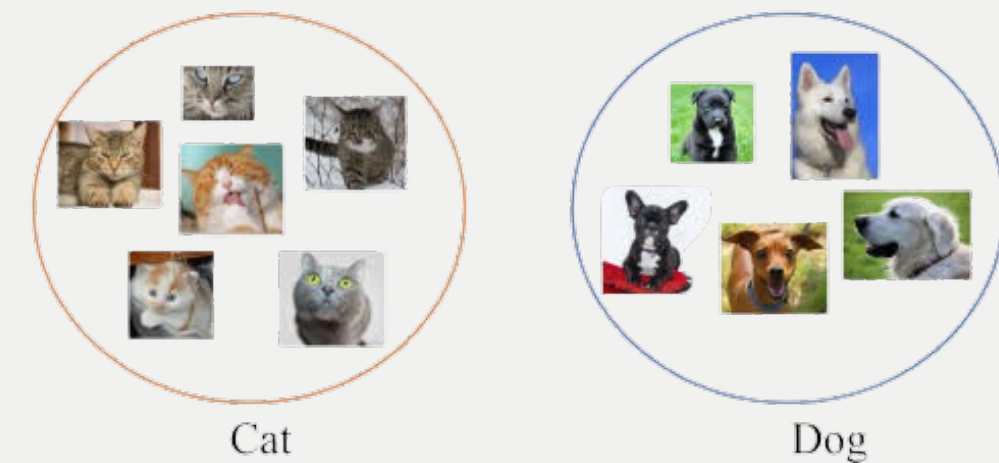
“Red stop and Green go for traffic lights”
Works for simple, fixed jobs.



WAY 2 · MACHINE LEARNING

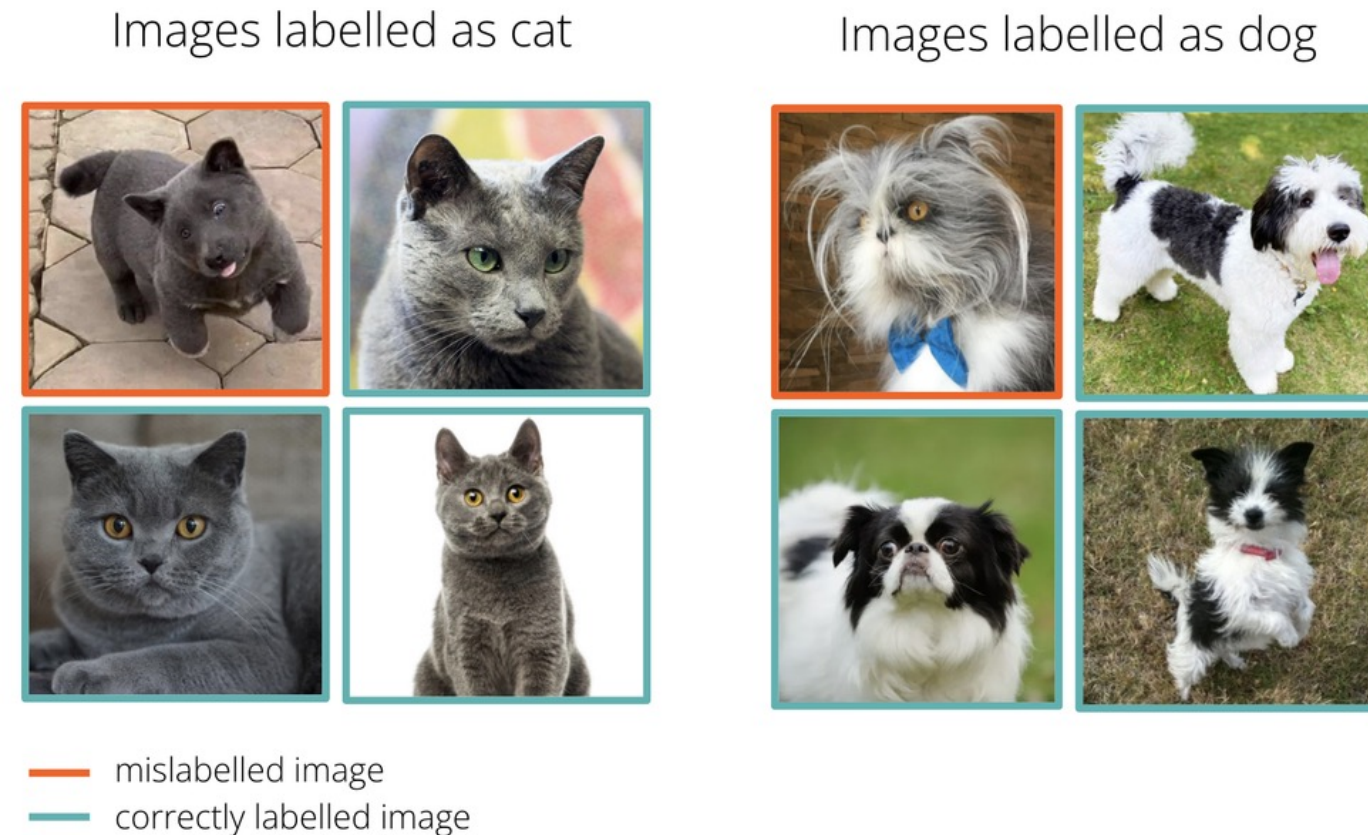
Show it examples with answers

How to tell apart cat and dog pictures, or other any animals or objects?
Can't write rules for all possible images.

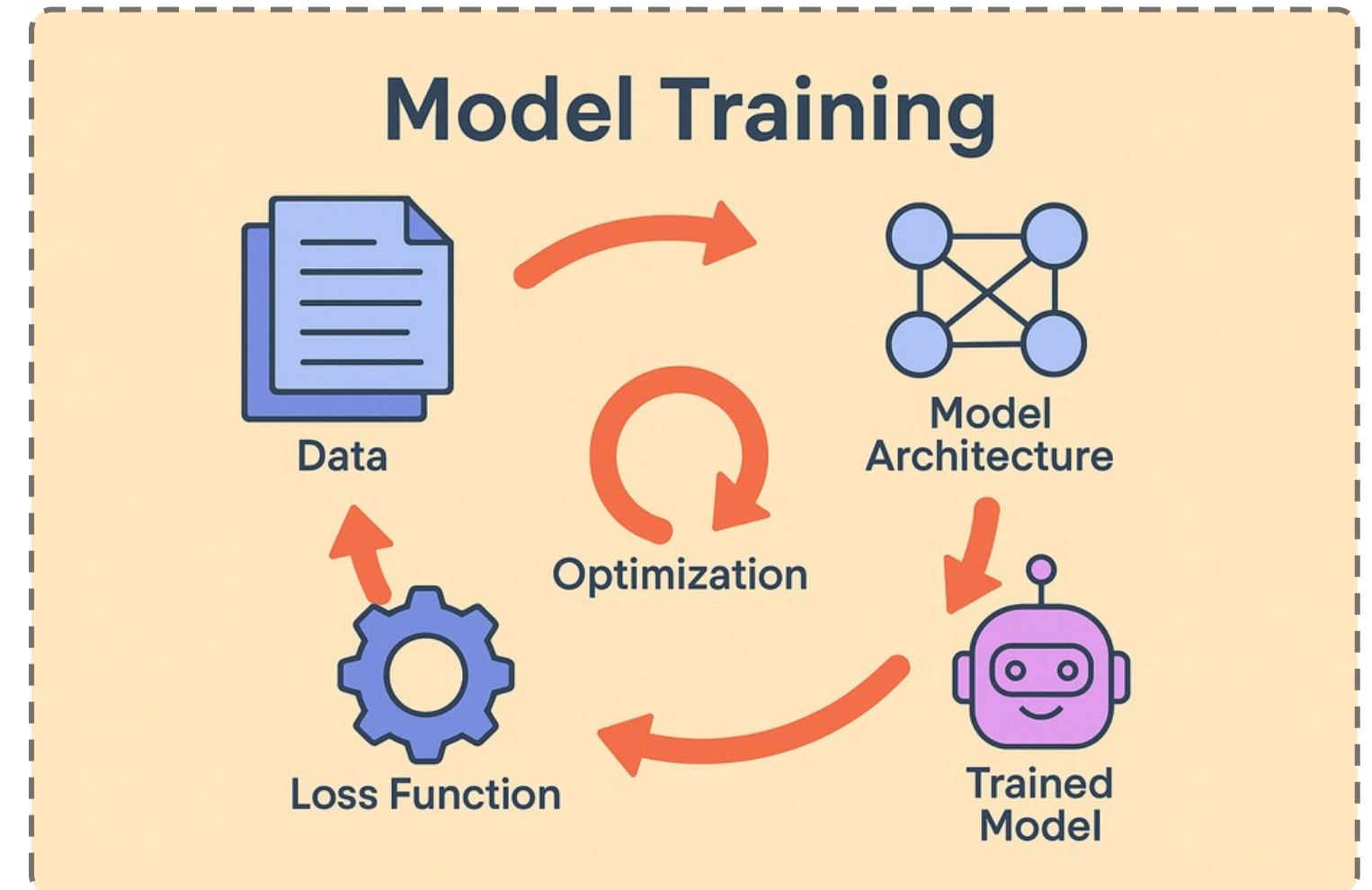


Model Training Process

1. Showing the AI many labelled examples - **Data**
e.g. one photo + its correct answer



1. The AI guesses → checks the answer → adjusts itself a little
2. Repeat hundreds of times → guesses get better



Remember — no rules written, AI **learns** through **data**
The smart brain after learning is called the **Trained Model**

Two datasets: training and test datasets

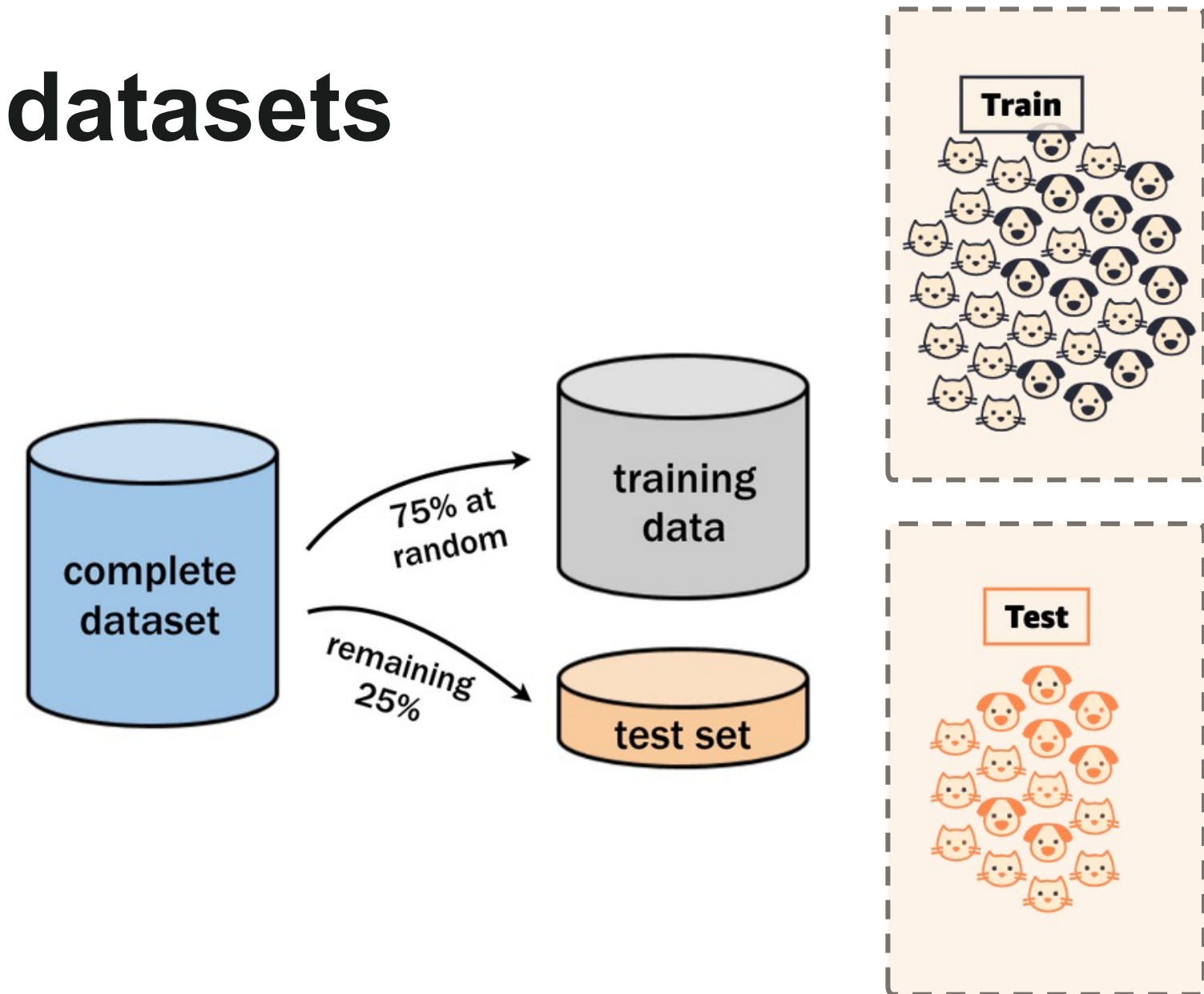
TRAINING DATA

Homework — with answers (correct label)

TESTING DATA

Exam — answers hidden (no label, just image)

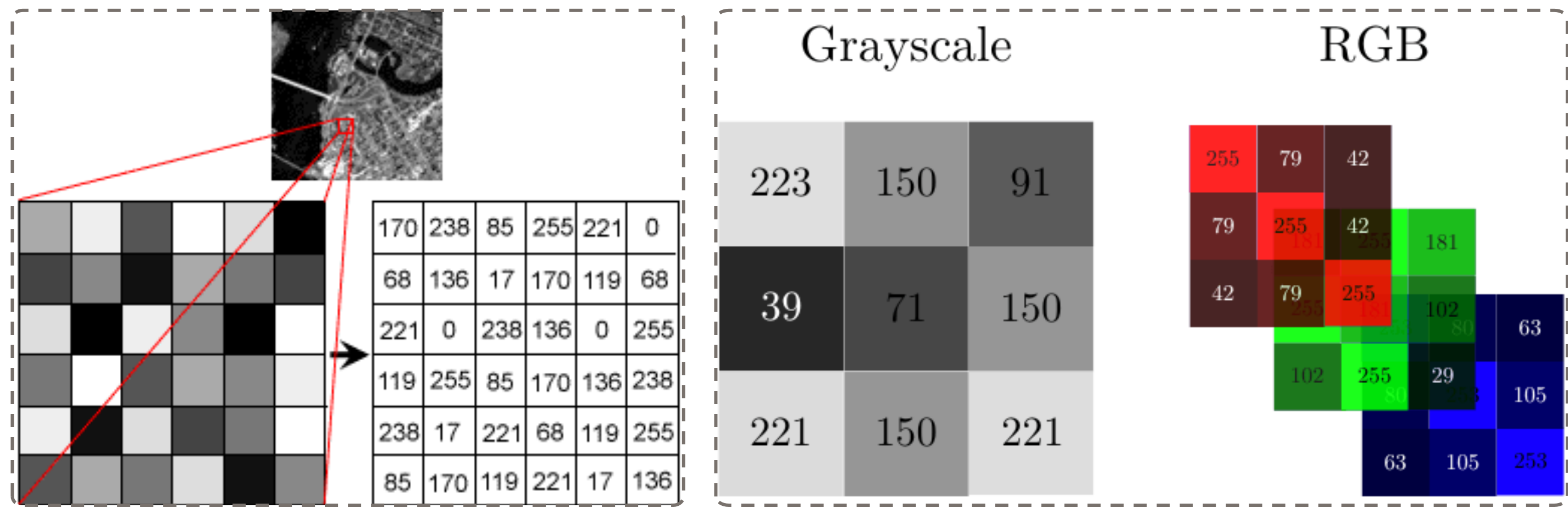
Never test on your training photos, that is like grading yourself on questions you already memorized.



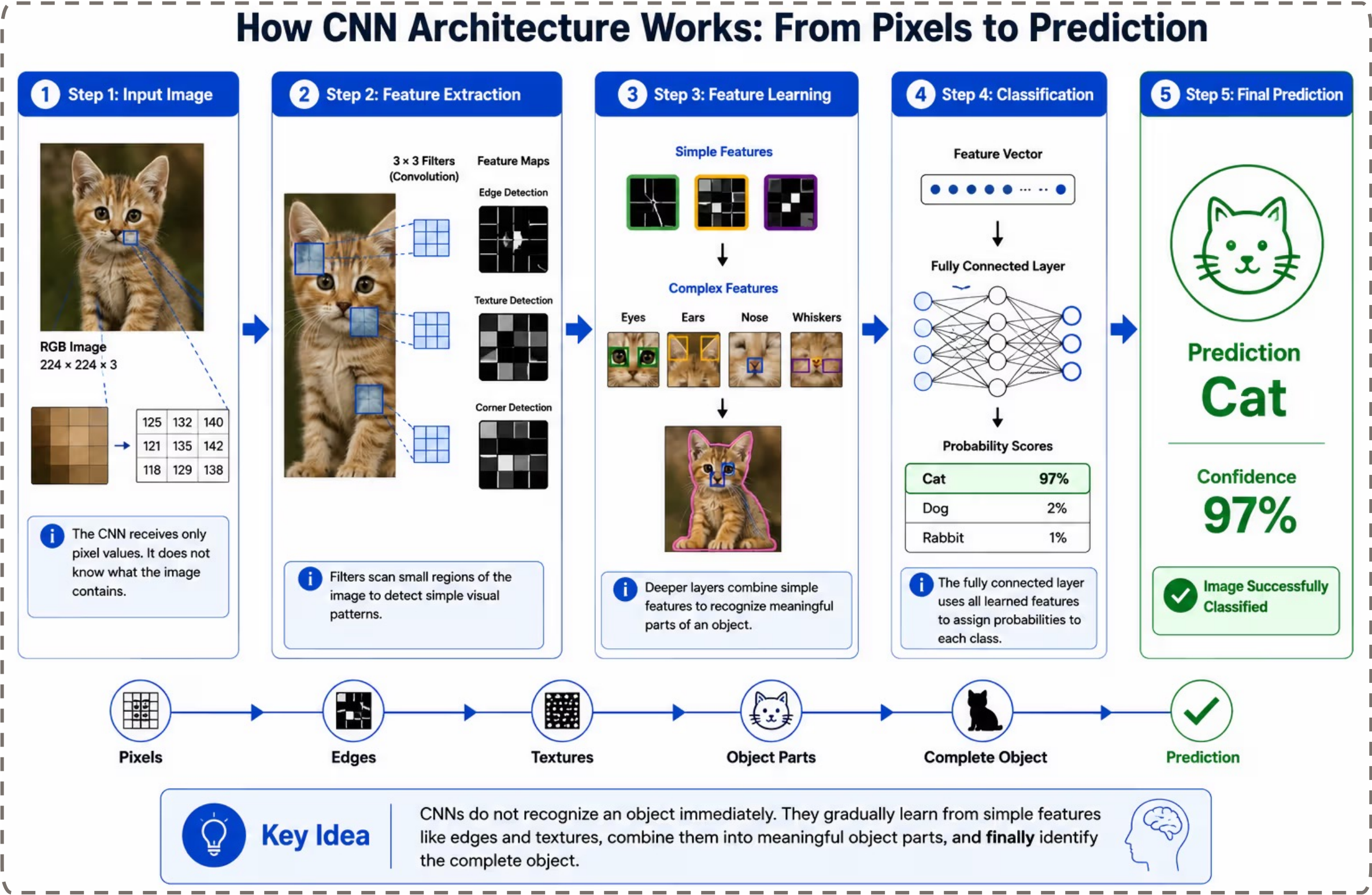
We usually use more training data (practice more) and less test data to verify the performance

Pixels into Digits

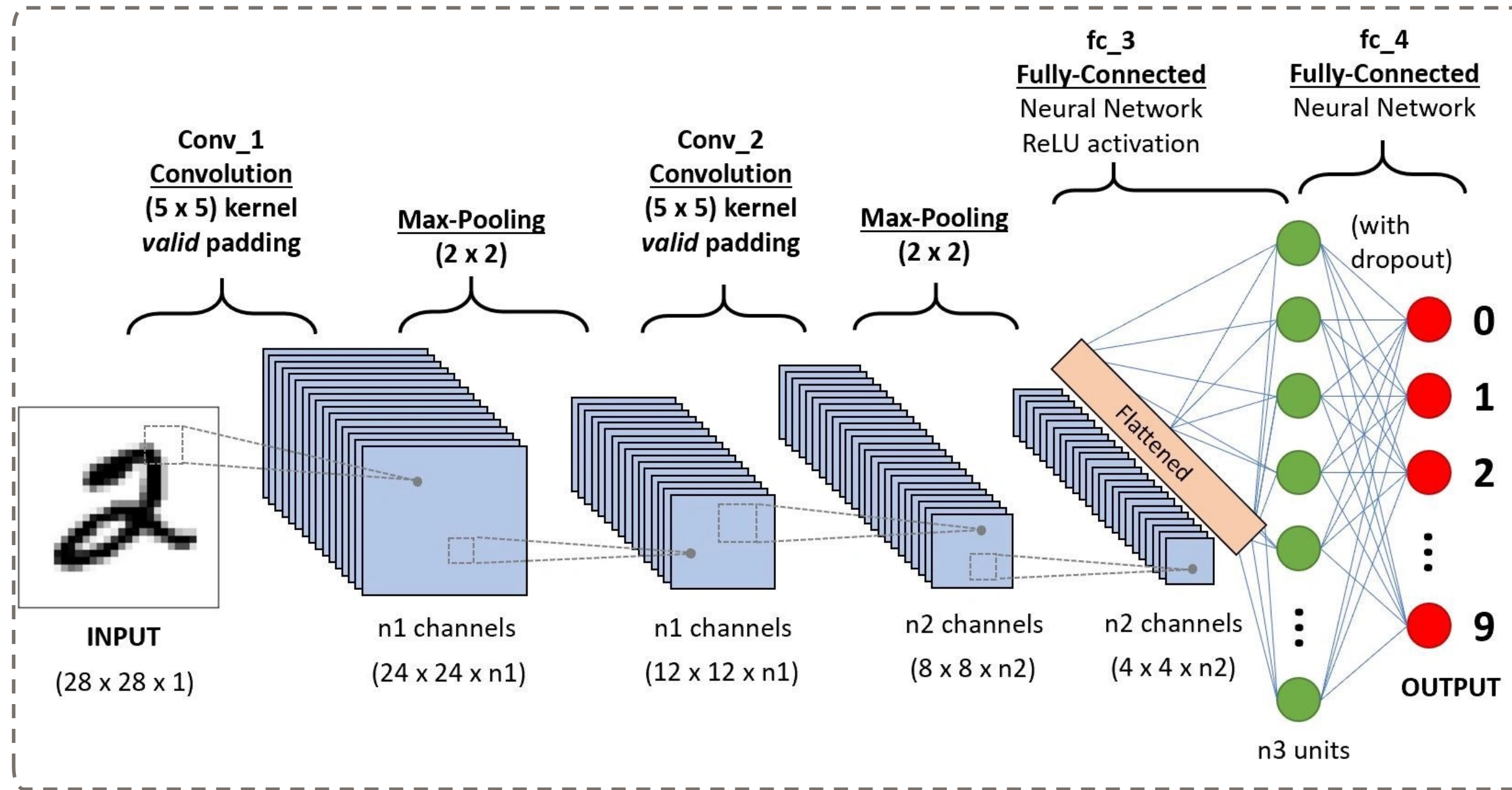
- A camera image is a grid of tiny squares — pixels
- Each pixel is one number: 0 = black, 255 = white, in between = different gray values
- Our model uses small 96×96 grayscale images — that is still 9,216 numbers
- Image recognition = finding patterns in those numbers



CNN (Convolutional Neural Network)



CNN (Convolutional Neural Network)



Confidence

how sure the model is, 0–100%

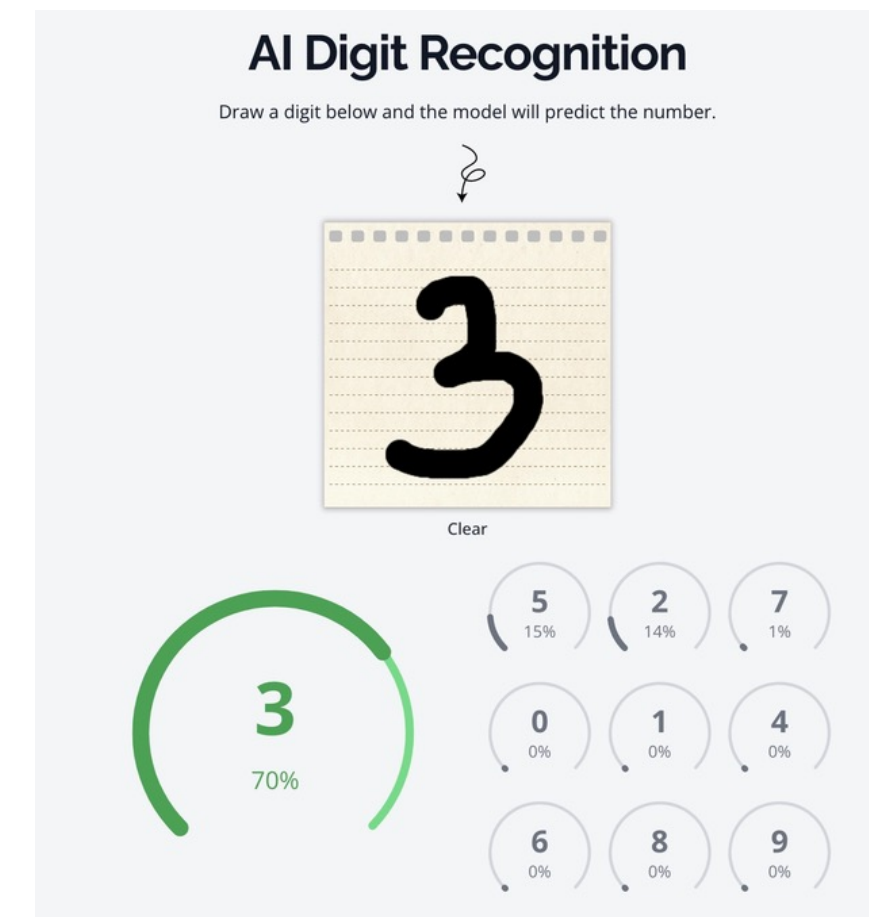
The CNN never outputs a plain yes or no. For every class/label it gives a confidence score.

For example:

The labels could be: Dog / Car / Apple



Or labels could be 1/2/3/4/5...9, numbers

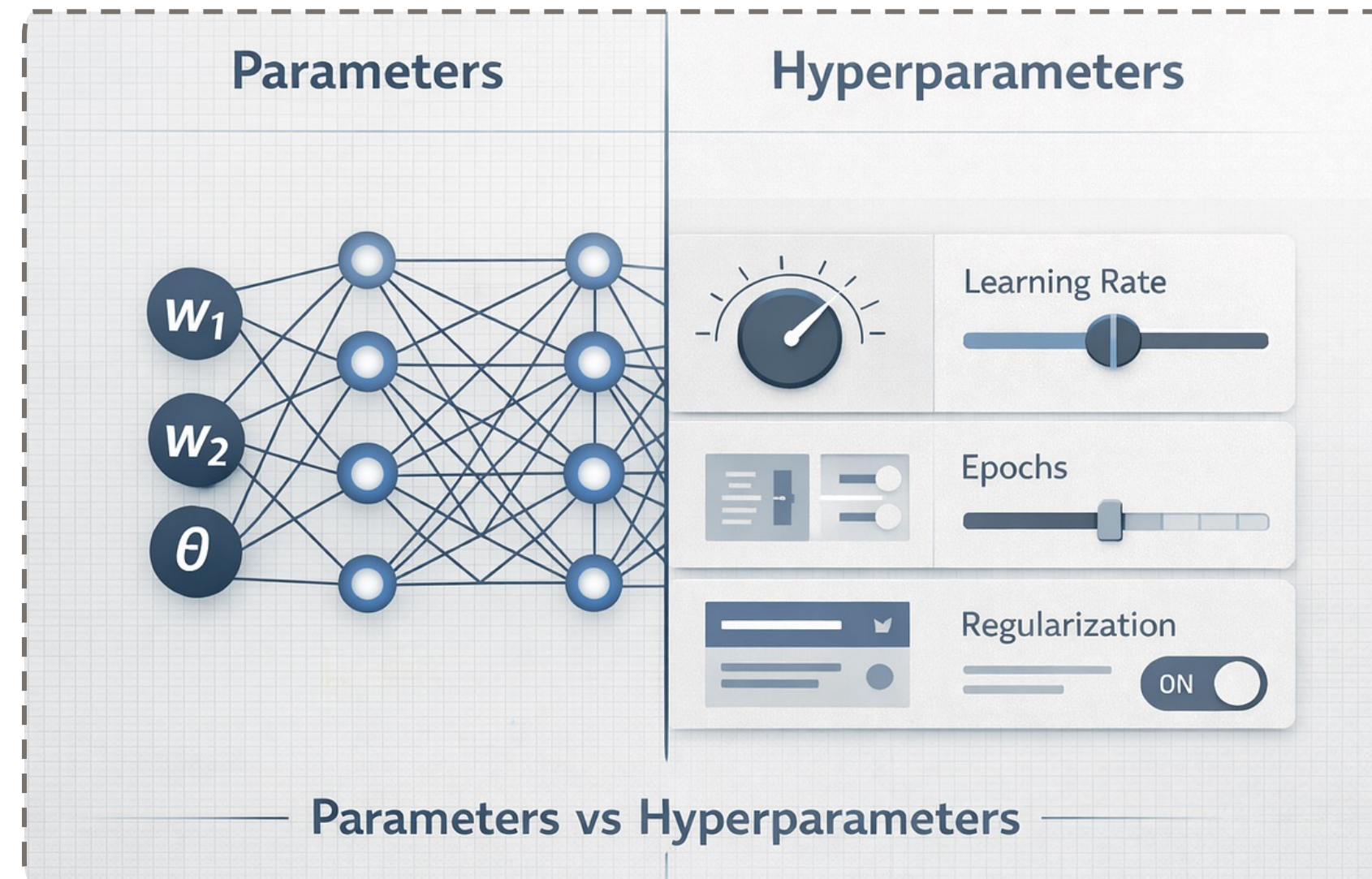


Careful — high confidence does not guarantee correct. The model can be **confidently** wrong !

Hyperparameters

PARAMETERS

Parameters are learned by the model, which you cannot change.



HYPERPARAMETERS

You choose the how it learns, the conditions: how long it studies, how fast it adjusts, etc.

Physical AI in the real world: the Tesla Cybercab



Tesla Cybercab | The Future is Autonomous
<https://www.youtube.com/watch?v=Qfj4urMF8CU>

Building Physical AI is not only an engineering question, but also a question of related to ethics.

- No driver, no steering wheel — cameras + AI make every decision
- Promised as a cheap robotaxi for everyone
- Same core idea as our car: camera in → AI decides → wheels act.

Questions every engineer must answer:

- How safe is “safe enough” to drive without a driver?
- If it crashes — who is responsible?
- What happens to the jobs of drivers?

LAB TUTORIAL

Road-sign recognizer.

1 · COLLECT

2 · PREPROCESS

3 · TRAIN

4 · TEST LIVE

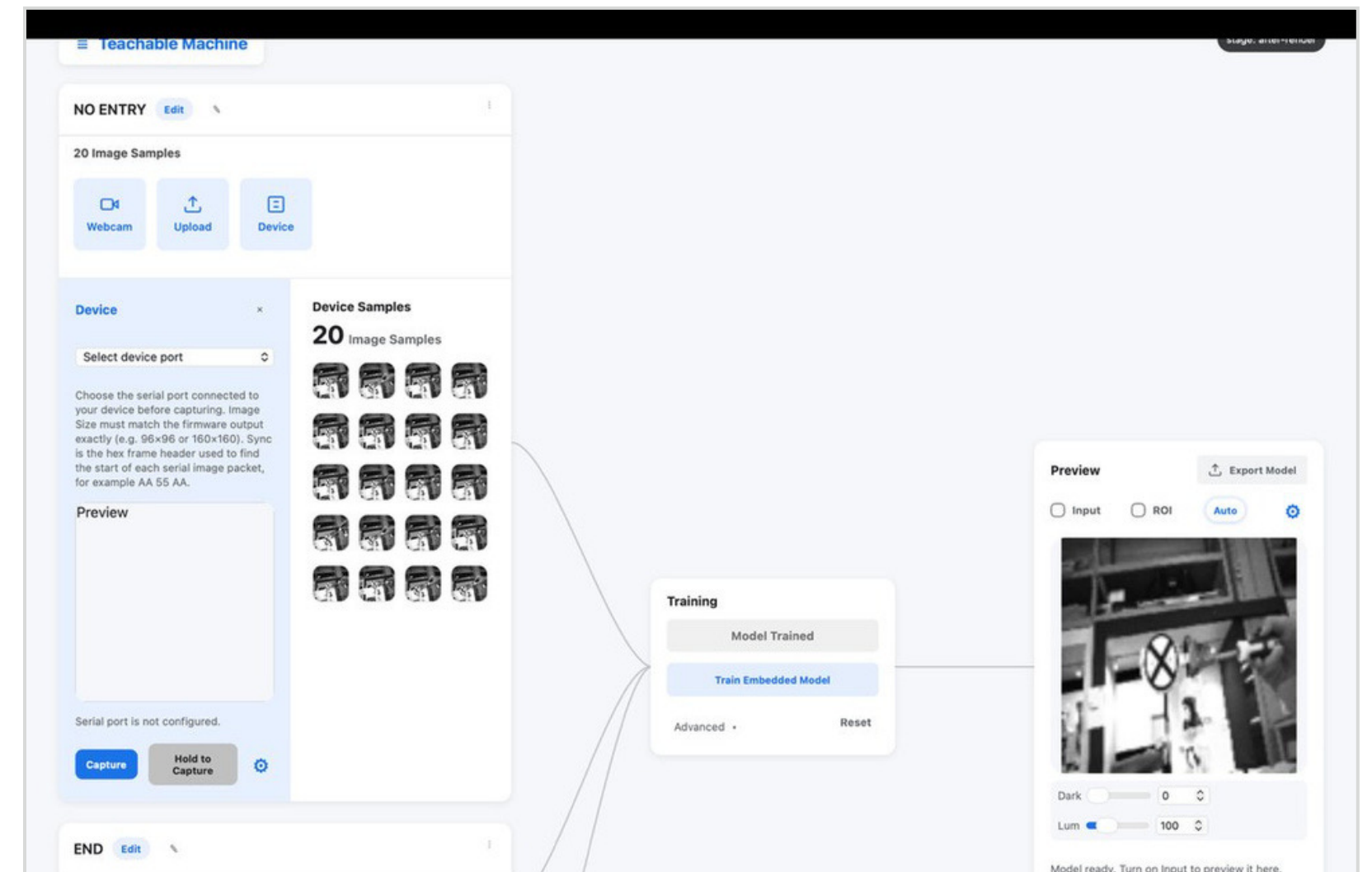
5 · EXPORT

6 · RUN ON CAR

Then you repeat the same steps with digits, in your lab time.

Collect the photos

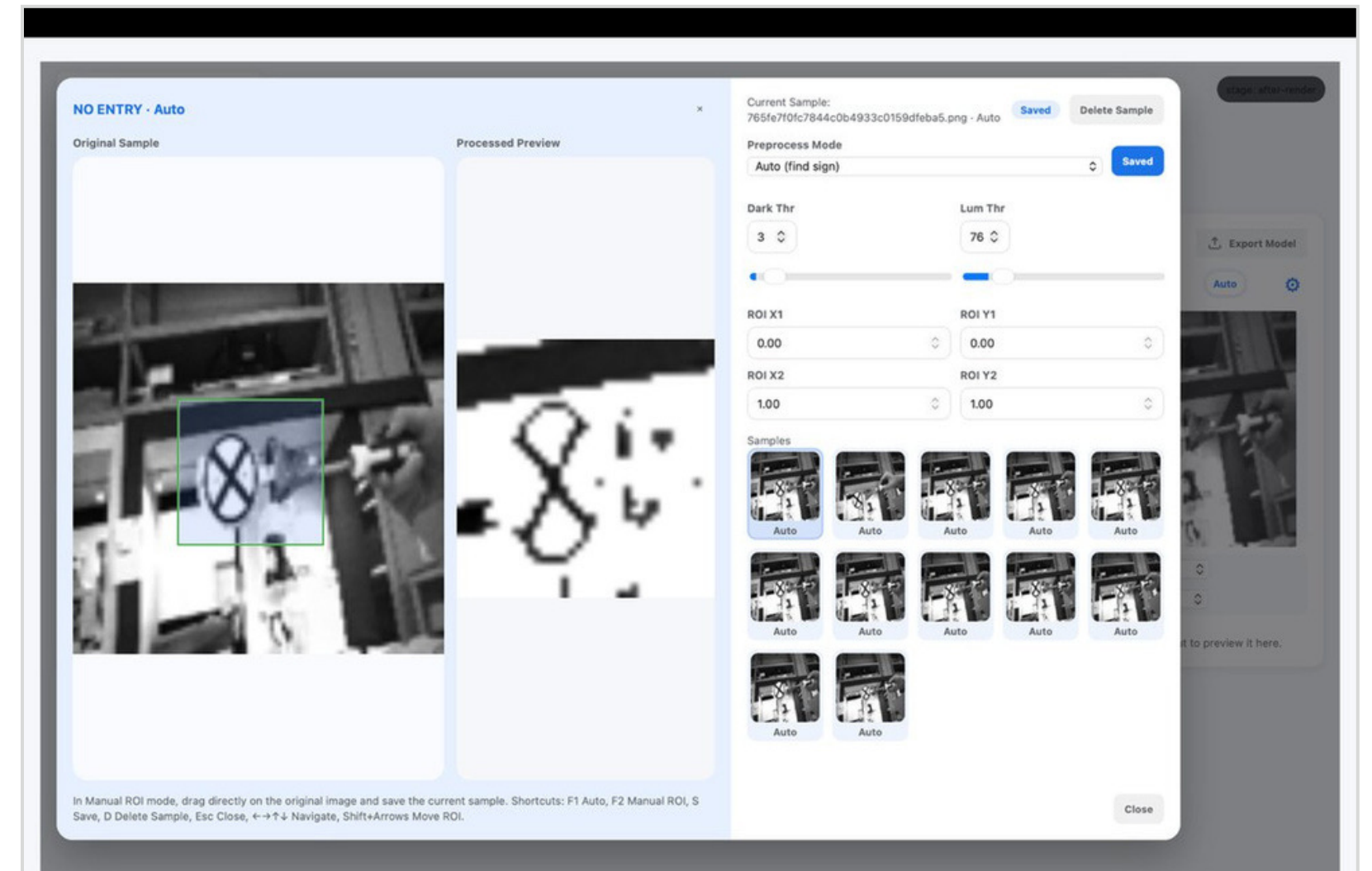
- Connect the car; in the app choose Device — photos come from the car's camera
- Three classes: left arrow · right arrow · stop
- ~40 photos per class, counts equal
- Vary it while shooting: near / far, straight / tilted, different sides of the frame



CAPTURING FROM THE CAR'S CAMERA

Clean the photos

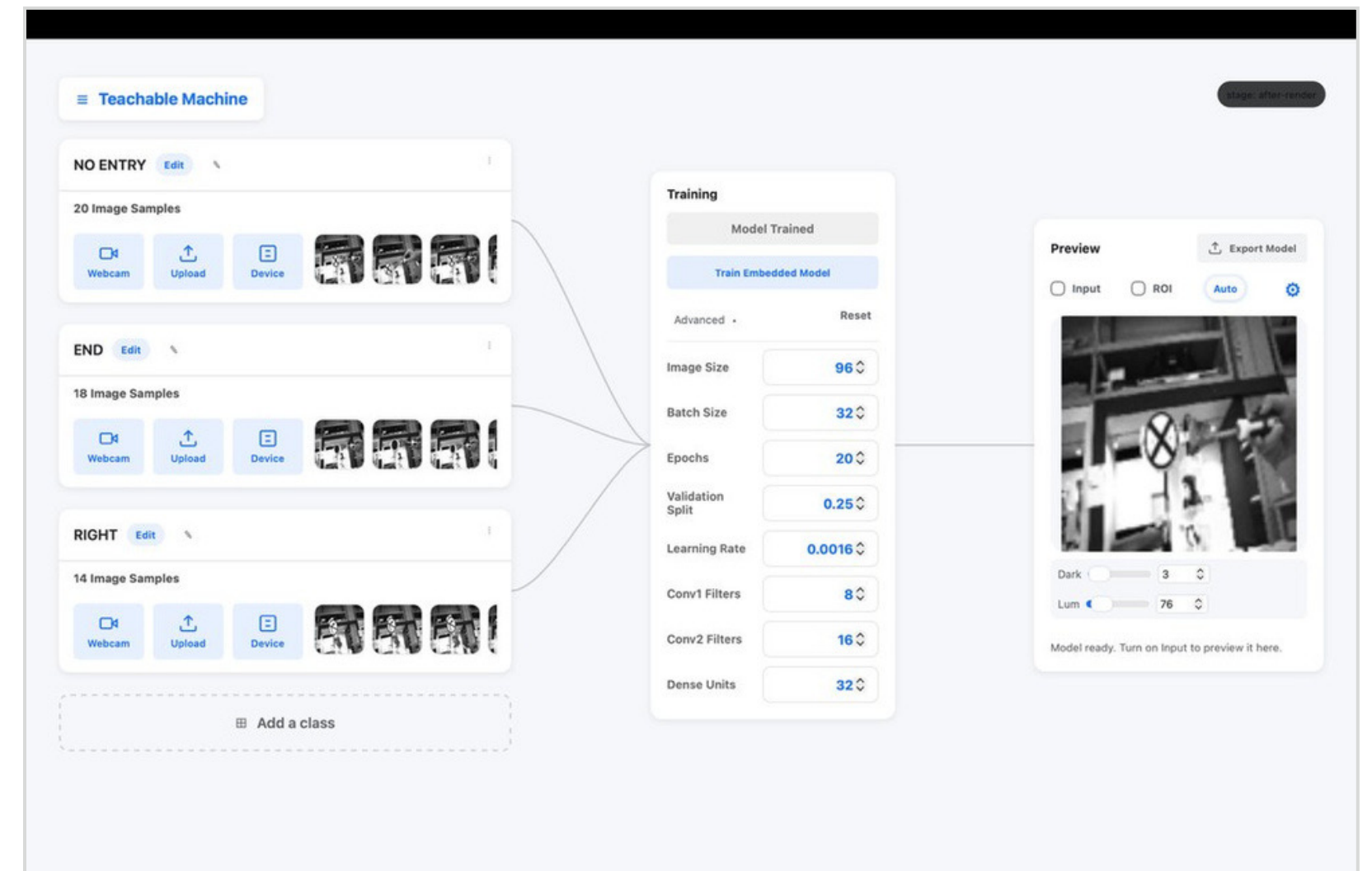
- Crop tight around the sign (the ROI box) — throw away the background
- Two thresholds clean the rest: too-dark pixels → white, too-bright pixels → white
- Target: black sign on white background
- Press S to save — every time



CROPPING AND THRESHOLDING

Train

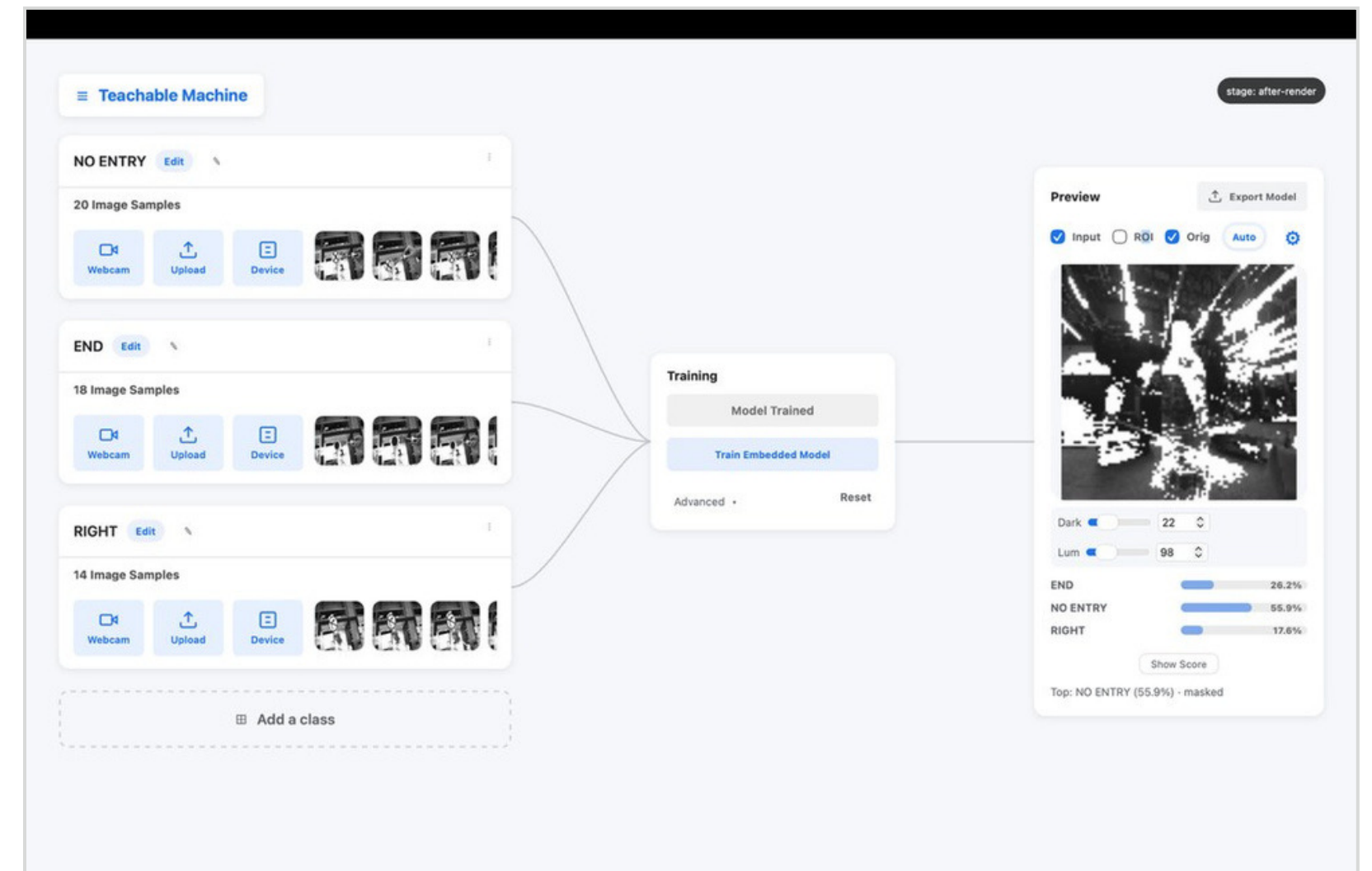
- Click “Use recommended” — the app fills in sensible settings
- Click “Train Embedded Model” and watch the accuracy climb
- Takes under a minute on this dataset
- If live results turn out poor later — the fix is usually more varied photos, not more training



THE TRAINING PANEL

Test it live

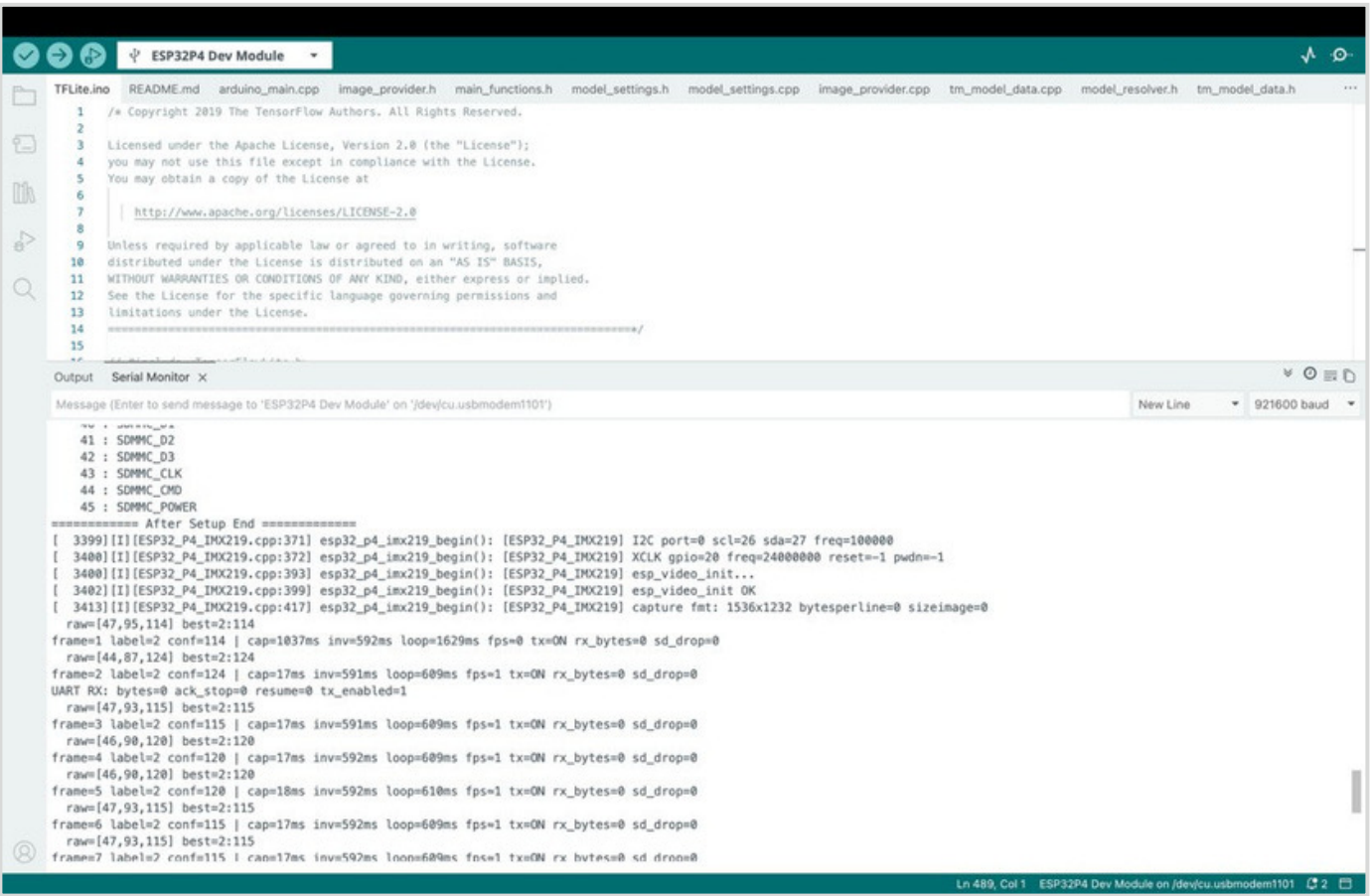
- Point at a sign — read the confidence bar
- Try a tilted sign, a far sign, a half-covered sign
- Show an empty scene: the model answers “No Sign” — the OOD gate says “no target here”, instead of guessing



LIVE CONFIDENCE + “NO SIGN”

Put it on the car

- Export → five small files
- Copy them into the project, check IMG_SIZE = 96
- Upload — the car now recognizes signs by itself
- Serial commands can adjust things later (get ood shows the settings)



THE MODEL RUNNING ON THE CAR

Your task: a digit recognizer

Train a model that reads handwritten digits 0–9 on paper, and runs live on the car.

RULE 1

Trained today

by your team, with photos you take here

RULE 2

Fair test

10 unseen digits written by ANOTHER team

RULE 3

Pass mark

8 of 10 correct = project complete

SHORTCUT

Short on time?

start with digits 1, 2, 3, 7 — the most confusable ones

Full instructions: check the lab manual

3 · HANDS-ON LAB · ~60 MINUTES

Build. Test. Ask.

Collect · 20 min

Preprocess · 10 min

Train · 5 min

Test · 10 min

Deploy · 10 min

Evaluate · 5 min

Stuck?

The lab manual has a troubleshooting table — port busy, one-class collapse, upload errors.

You can always ask the TA for help at anytime

Better result?

Train → test → fix the photos → retrain.
Two or three loops are normal.

Finished early?

Extensions in the manual: fewer photos, bigger model, new handwriting.

Demo & sharing

Every team shows what they built — 1 minutes each.

SHOW

a live demo: your model reading digits on the car

SHOW

your scorecard: the 10 unseen digits and your result

TELL

one change you made, and what happened

ASK

what would you try with one more hour?

Thank You

Q&A